

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I K.R.Jeevan Balaji(192471032) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. R. Jeevan Balaji

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1. Learning Agent for Personalized Education System



1. Introduction

A Learning Agent for Personalized Education System is an intelligent software agent that observes a student's learning activities, analyzes their performance, and provides suitable learning content based on their individual needs. Unlike traditional education systems that provide the same content to all students, a learning agent continuously adapts the learning experience according to the student's performance, strengths, weaknesses, and learning speed.

The main objective is to improve learning performance, personalize education, adapt content difficulty, provide timely feedback, and help students learn effectively at their own pace.

2. What is a Learning Agent?

A Learning Agent is an intelligent agent that:

- Observes the student's learning behavior.
- Collects performance and assessment data.
- Analyzes strengths and weaknesses.
- Adapts the difficulty of learning content.
- Provides personalized recommendations.
- Learns and improves its decisions using feedback.

For example:

IF a student consistently answers advanced questions correctly
THEN increase the difficulty level and recommend advanced learning materials.

IF a student repeatedly makes mistakes in a particular topic
THEN provide simpler explanations and additional practice questions.

3. System Components

Student Data

- Quiz and examination scores
- Answers and mistakes
- Time taken to complete questions
- Learning activity and interaction history
- Completed lessons and assignments

Learning Agent

- Receives student learning information.
- Analyzes the student's current performance.
- Identifies strengths and weaknesses.
- Determines the appropriate difficulty level.
- Recommends personalized learning content.

Student Model

- Maintains the student's knowledge level.
- Stores performance information.
- Identifies weak and strong subjects.
- Tracks learning progress over time.

Feedback System

- Collects feedback from quizzes and activities.
- Measures whether the student has understood the topic.
- Updates the student model based on performance.
- Helps the agent improve future recommendations.

Content Recommendation System

- Recommends suitable lessons and videos.
- Provides practice questions.
- Selects content according to the student's ability.
- Suggests advanced content when performance improves.

PySpark Processing System

- Processes large amounts of student learning data.
- Analyzes data from multiple students simultaneously.
- Identifies common learning patterns and difficulties.
- Supports large-scale machine learning and recommendation tasks.
- Provides processed information to the Learning Agent.

Communication System

- Transfers information between the student platform, Learning Agent, and data-processing system.
- Allows the agent to receive updated performance data.
- Delivers personalized recommendations and feedback to students.